Robot dance studio

Loops in motion + a cone pusher

Today’s robot challenge

A short two-action routine

A repeat count you can explain

A clear space for the whole dance

GREEN steady: on **ORANGE flashing: charge** **RED flashing: stop + mentor**

Center button lights. A letter on the display is your program’s output.

Counted loops and indentation

```
for beat in range(3):
    await drive(90)
    await turn(90)
await signal("D")
```

Our prediction

How many drives happen in three repeats?

Does a signal outside the loop repeat?

Reading fragment from the complete supplied project

Our robot and build jobs

Robot team: predict, run, count. Rotate jobs.

Build team: make a removable front cone pusher.

Try 3 hand pushes. Change one feature. Try 3 more.

Goal: upright cone, whole base inside the zone.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Build a cone pusher

1. Build a low front bar with two secure mounts. Use the proxy.
2. Keep the bar off the floor and clear of the wheel outline.
3. Hand-push 10 cm. Try 3 times from the same start.
4. Change ONE: height, width, bracing or a shallow V.
5. Try 3 more times. Compare successes out of 3.

Success: cone stays upright; its whole base finishes inside the zone.

Rotate builder, designer and tester. Mentor checks fit before robot testing.

Evidence from our test

```
for push in range(3):
    await drive(60)
    await pause(300)
await signal("D")
```

Keep it upright.
Whole base in zone.
Stop if it tips or escapes.

Predict: moves, pauses, signals?

What did our design change improve?

Cone retest: mentor checks fit and target. No turns. Keep SPEED = 200.

What we learned and noticed

What did you learn about repeating actions?

What did you notice during the robot dance?

What would you change in the loop or cone pusher?

Save your team project. Store your build. Reset the robot and parts.